Computational Genomics (76553) HUJI

Yes. Here is the clean story of what happened.

1. What we were trying to prove

The original biological question was:

> In MASLD progression, do hepatocytes lose liver zonation?

"Zonation" means that hepatocytes are not all equivalent. In a healthy liver lobule, cells near the **central vein** express one program, and cells near the **portal side** express another. A healthy spatial liver atlas is useful here because liver lobules are physically polarized by portal-to-central blood flow, creating oxygen/nutrient gradients and spatially segregated hepatocyte functions.

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The old sentence was something like:

> "The pericentral program turns off at end-stage."

That sentence is now too crude and probably wrong.

2. What we first tried, and why we became suspicious

The dataset from Paper 1 is **spatially blind snRNA-seq**. We have nuclei and gene expression, but we do **not** know where each nucleus physically sat in the lobule.

So the assistant used a healthy spatial atlas as a **ruler**: pericentral markers on one side, periportal markers on the other, then infer a pseudo-zonation coordinate for each hepatocyte.

That was useful for exploration, but it created danger:

A pseudo-coordinate built from marker genes can become circular. If the genes used to define "pericentral" change in disease, then the coordinate itself changes. Then asking "does gene X lose its zonation slope along this coordinate?" can mix up three things:

1. actual spatial identity,
2. total gene expression level,
3. behavior of the ruler itself.

So we started rejecting metrics like slopes, correlations, z-scores, IQR/spread, and log fold-change as final evidence. They summarize patterns, but they do not answer the concrete biological question:

> Which genes changed? In which cells? In how many donor samples? In how many nuclei? Using actual molecules or model-corrected values?

That reset the logic.

3. The major confound we discovered

The stages were **not sampled the same way**.

From the methods: needle biopsies were taken with ultrasound guidance, but healthy donor and explant tissue were handled as approximately 1 cm^3 cubes. Also, for two healthy donors and all end-stage patients, samples were taken from left, right, and caudate lobes.

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So this is the key design problem:

NAFLD/NASH/cirrhosis: right-lobe needle biopsy
End-stage: transplant explant, multiple lobes

Therefore, “end-stage” is tangled with:

- explant tissue,
- surgery/transplant context,
- possible ischemia/handling stress,
- multi-lobe sampling,
- different tissue geometry and processing context.

This does **not** mean the data are useless. It means the sentence “end-stage disease causes X” is not automatically identifiable from this cohort.

4. The lobe finding

The assistant checked end-stage samples across caudate/left/right lobes and found that the pattern looked similar across lobes.

That matters because it weakens one confound:

> The end-stage pattern is probably not simply caused by mixing different lobes.

But it does **not** remove the bigger confound:

> All end-stage samples are still explants.

So the corrected conclusion is:

multi-lobe sampling confound: mostly reduced
explant/ischemia/handling confound: still alive

And yes, the main disease-stage analysis should still always be **right-lobe-only first**, because that is the most comparable subset.

5. What the exploratory SCT analysis found

Using the processed matrix, the assistant found:

A. CYP2E1 / CYP1A2 / AKR1D1 / ADH4 / SLCO1B3 drop mainly at end-stage.

This suggested that a specific module collapses: not all pericentral identity, but a xenobiotic/alcohol/redox/transport subprogram.

B. Periportal genes rise at end-stage.

PCK1 detection rises strongly; ALDOB stays broadly detected and its level rises. This suggests periportal-like metabolic induction, but not necessarily true expansion of periportal cells. That still needs a cell census.

C. GLUL and CYP3A4 do not collapse.

This is what killed the old story. GLUL is a canonical pericentral identity gene. CYP3A4 is also a major liver xenobiotic gene. If the whole pericentral program turned off, these should drop too. They do not.

So the better exploratory story became:

> Not “pericentral turns off,” but “some pericentral identity remains, while a selective detox/redox/transport submodule drops.”

D. Stress genes turn on only at end-stage.

FOS, JUN, HSPA1A and similar stress genes are almost absent in biopsy stages and appear at end-stage. That is exactly why the explant/ischemia concern became serious.

Independent liver atlas work explicitly warns that tissue source and ischemia can alter liver expression, and even says post-mortem ischemia can cause transcript-dependent RNA degradation explaining a large fraction of RNA-seq variability.

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So the fact that stress genes appear at the same stage where detox genes collapse is a major warning.

6. Why the SCT matrix is not good enough

This is the core point.

The assistant computed “LEVEL” as:

$$\log(1 + 10{,}000 \times \frac{\text{gene value in cell}}{\text{total cell values}})$$

If the input were **raw UMI counts**, this would roughly mean:

> Out of all molecules captured in this nucleus, what fraction came from this gene?

That would be interpretable.

But the input was not raw UMI counts. It was **SCTransform-corrected/model-corrected data**.

SCTransform is designed to remove technical variation, especially sequencing depth effects. That is useful for clustering and visualization. But after this transformation, the numbers are no longer literal molecule counts. They are model-adjusted expression values.

So if we then compute “counts per 10,000” from SCT values, we are doing something conceptually wrong:

> We are pretending model-corrected values are molecules.

They are not.

This creates two problems.

First, the **LEVEL scale is not physical**. A LEVEL of 0.13 does not mean a real abundance of 0.13 molecules, or 0.13 per 10k molecules, or anything biologically direct.

Second, even **fraction expressing** becomes suspicious if “expressing” means “SCT value > 0.” In raw counts, count > 0 means “we captured at least one molecule.” In SCT space, value > 0 may mean something about model-corrected residual/corrected expression, depending on the slot. That is not the same physical event.

7. How we empirically proved SCT is bad for this purpose

The key sanity check was **ACTB**.

ACTB is a housekeeping gene. It should be broadly and robustly detected in hepatocyte nuclei.

But in the assistant’s LEVEL scale, ACTB was around **0.13**.

Back-transforming:

$$e^{0.13} - 1 \approx 0.14$$

So if we interpreted the LEVEL literally, ACTB would look like roughly **0.14 counts per 10,000**, which is biologically absurd for a housekeeping gene.

That is not a subtle statistical objection. It is an empirical proof that the LEVEL scale is not an absolute expression scale.

So we learned:

SCT LEVEL values cannot be used to say how much expression exists.
They can maybe suggest directions, but they cannot support final quantitative claims.

The current SCT findings are therefore **exploratory diagnostics**, not final results.

8. Why raw counts solve the specific problem

Raw counts are not perfect, but they have a physical meaning.

For a donor, gene, and cell set:

$$N = \text{number of hepatocyte nuclei}$$

$$k = \text{number of nuclei with raw UMI count} > 0$$

$$k/N = \text{fraction of nuclei where we captured at least one molecule}$$

$$M = \text{total raw UMIs for that gene}$$

$$E = \text{total raw UMIs across all genes}$$

$$10{,}000 \times M/E = \text{raw gene molecules per 10k captured molecules}$$

Now every number has a physical interpretation.

Example:

> CYP2E1 raw-positive = 49% means 49 out of 100 hepatocyte nuclei had at least one captured CYP2E1 molecule.

That is much better than:

> CYP2E1 SCT-positive = 49%, whose exact meaning depends on model-corrected values.

Raw counts also let us separate different mechanisms:

Cell depletion: fewer PC-like hepatocytes exist.
Gene turnoff: PC-like hepatocytes still exist, but CYP2E1 disappears inside them.
De-zonation: PC and PP markers co-occur in the same nuclei.
Stress artifact: detox collapse occurs mostly in FOS/JUN/HSPA1A-high cells.

SCT values blur these distinctions.

9. The methodological reset

The reset was:

Stop asking:

> What happens to a normalized zonation score?

Start asking:

> How many nuclei of each interpretable type exist per donor, and how many raw molecules of each gene are detected?

So the next correct primitives are:

1. **Raw-count donor-level detection fractions**
   For each gene: how many hepatocyte nuclei are raw-positive?
2. **Raw-count molecule burden**
   How many raw UMIs for this gene per 10k total hepatocyte UMIs?
3. **Anchor-cell census**
   Count PC-anchor, PP-anchor, dual, and null hepatocytes.
4. **Stress stratification**
   Compare detox genes in stress-high versus stress-low hepatocytes/samples.
5. **Right-lobe-only as the main disease-stage comparison**
   Multi-lobe only as a secondary end-stage internal check.

10. Current honest status

What we know now:

- The old "pericentral program turns off" story is not supported.
- The processed data suggest selective loss of **CYP2E1/CYP1A2/ADH4/AKR1D1/SLCO1B3**, not wholesale pericentral loss.
- GLUL and CYP3A4 being retained/rising is the key contradiction.
- Periportal genes, especially PCK1, appear induced at end-stage.
- Stress genes turn on specifically at end-stage.
- Lobe mixing probably does not explain the pattern.
- Explant/ischemia/handling stress still might explain part or all of the dramatic end-stage step.
- The current SCT-based percentages and levels are not final evidence.
- The next analysis must fall back to raw RNA counts.

The cleanest summary sentence is:

> We reset the analysis because SCT-normalized values failed an absolute-expression sanity check and because the end-stage signal is structurally confounded by explant/stress; the processed data still usefully show that "pericentral shutdown" is the wrong model, but the defensible next step is raw-count, donor-level counting of PC/PP/stress/detox states, especially in the right-lobe-only set.

can we justify why we were confounded by that? shouldn't we have a-priori understand all of this from reading the papers carefully?

Yes — partly. A careful methods-first read **should have flagged this earlier**. There is no need to pretend otherwise.

But there is also a reasonable explanation for why it was missed: the paper's scientific narrative is organized as **healthy → MASLD → MASH → cirrhosis → end-stage**, so it invites you to treat stage as the main axis. The confound is not in the headline result; it is in the acquisition details: needle biopsies versus healthy/explant cubes, and multi-lobe sampling for end-stage. The paper describes 47 biopsies across disease stages in the main text, but the methods reveal that ultrasound-guided needle biopsies and explant/healthy donor tissue were collected differently, and that all end-stage patients contributed left/right/caudate samples.

So the honest version is:

> We should have built the sample-design table before interpreting biology. We did not. The confound was discoverable from the paper, but it was easy to miss because the paper's biological framing treats progression stage as the dominant axis.

That is a **justification**, not an excuse.

What we should have done a priori

Before any zonation score, UMAP, slope, correlation, or gene table, we should have made a table like:

donor	stage	tissue source	lobe	biopsy/explant/surgery	batch	processing	N hepatc

Then ask:

> Is disease stage confounded with tissue source, lobe, batch, processing, or QC?

If the answer is yes, then the analysis has to be designed around that from the start.

For this dataset, the danger is severe because:

milder disease = right-lobe biopsy
end-stage = explant, multiple lobes

So "end-stage" is not just a biological label. It also means "different acquisition regime."

Why the mistake was understandable

Several reasons.

First, the paper itself emphasizes disease progression. Its abstract says hepatocytes lose zonation and that liver architecture changes during MASLD progression. That frames the reader to think "stage effect."

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Second, “biopsy” is used broadly in the biological narrative. But experimentally, a needle-core biopsy and a cube cut from explant tissue are not equivalent. They differ in size, handling, ischemia exposure, spatial composition, and possibly processing delay.

Third, the end-stage biology is genuinely dramatic. When you see huge shifts in CYP2E1, CYP1A2, AKR1D1, PCK1, FOS, JUN, HSPA1A, it is tempting to read them as disease progression. But the exact same step is also where the sampling regime changes. That is the trap.

Fourth, the raw/SCT issue is not something the paper necessarily tells you clearly. It depends on the object the assistant used. If the assistant silently used an SCT assay instead of raw RNA counts, you only discover the problem by auditing the data object and sanity-checking values like ACTB.

What we learned from this

The reset is methodological:

We stopped trusting “stage” as a clean biological variable until the design table proves it is clean.

We also stopped trusting normalized abstract summaries as final evidence. Instead of asking:

> Does the zonation score collapse?

we now ask:

> In each donor, how many raw-count hepatocyte nuclei express each gene? How many PC-anchor, PP-anchor, dual, and null nuclei exist? Does the result survive right-lobe-only? Does it track stress genes?

That is much more defensible.

How to phrase this without sounding incompetent

You can say:

> The original analysis followed the paper’s biological framing of disease progression, but a later audit of sample provenance showed that the end-stage category is structurally confounded with explant acquisition and multi-lobe sampling. We therefore reset the analysis around sample source, lobe, raw-count molecule detection, and stress controls before making any disease-stage claim.

That is honest and professional.

The sharper lesson

This was not merely “we missed a detail.”

It was a hierarchy-of-evidence problem.

We began with:

1. biological story,
2. marker genes,
3. pseudo-zonation,
4. normalized summaries.

But the correct order is:

1. sample design,
2. raw data scale,
3. donor-level counts,
4. confound checks,
5. then biology.

So yes: a careful a-priori read should have caught it. But the important part is that once we caught it, we did the right thing: downgraded the old interpretation, treated SCT results as exploratory, and rebuilt the analysis around raw counts and structural confounds.

Sources